Inference often combines conditioning and summarizing

Observed y $\rightarrow$ **Condition** $p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$ $\rightarrow$ **Summarize** $\hat{z} = E[z \mid y] = \mu_{z|y}$

Murphy, Book 2 §2.3.2.2–2.3.2.3

$$E[z \mid y] = \int z p(z \mid y) \, \mathrm{d}z$$

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z|y}$

A point estimate needs a loss function

Carried forward Inference often combines conditioning and summarizing

$$\hat{z} = E[z|y] = \mu_{z|y}$$

Observed y; unknown $z \in \mathbb{R}^d$; $a \in \mathbb{R}^d$ fixed at this y.

Squared-loss criterion

$$\text{Assume } E[\|z\|^2|y] < \infty$$

$$L(a, z) = \|z - a\|^2, \quad R(a|y) = E[L(a, z)|y]$$

Center at m

$$m = E[z|y], \quad V = \text{Cov}[z|y], \quad r = z - m$$

Earlier Gaussian: $m = \mu_{z|y}$, $V = \Sigma_{z|y}$.

$$z - a = r + (m - a)$$

$$\|z - a\|^2 = \|r\|^2 + 2r^T \cdot (m - a) + \|m - a\|^2$$

Murphy, Book 1 §5.1.5.1; vector decomposition: course elaboration

- We choose a to reconstruct z by minimizing posterior expected squared error $R(a|y)$.
- Centering writes the error as random residual r plus the fixed offset $m - a$.

The conditional mean uniquely minimizes squared-error risk

Carried forward Previous slide: m, V, and the pointwise squared-error expansion

$$m = E[z|y], \quad V = \mathrm{Cov}[z|y], \quad r = z - m$$
$$\|z - a\|^2 = \|r\|^2 + 2r^T \cdot (m - a) + \|m - a\|^2$$

Take conditional expectation; a, m are fixed at this y.

$$R(a|y) = E\left[\|r\|^2|y\right] + 2E\left[r^T \cdot (m - a)|y\right] + \|m - a\|^2$$

Centering cancels the cross-term

$$E[r|y] = E[z|y] - m = 0$$
$$E\left[r^T \cdot (m - a)|y\right] = E[r|y]^T \cdot (m - a) = 0$$

Residual energy; $V = E[rr^T|y]$

$$E\left[\|r\|^2|y\right] = \sum_{j=1}^{d} E\left[r_j^2|y\right] = \sum_{j=1}^{d} V_{jj} = \mathrm{tr}\, V$$

$$R(a|y) = \mathrm{tr}\, V + \|m - a\|^2$$

Substitute the two moments.

$$\|m - a\|^2 \geq 0 \quad \text{with equality iff} \quad a = m$$
$$a^*(y) = m = E[z|y], \quad \min_a R(a|y) = \mathrm{tr}\, V$$
$$\text{Earlier Gaussian:} \quad a^*(y) = m = \hat{z} = \mu_{z|y}$$
$$\min_a R(a|y) = \mathrm{tr}\, \Sigma_{z|y}$$

- Risk is residual uncertainty plus the squared distance of the chosen action from the mean.
- This is model expected loss; finite conditional second moments suffice without Gaussianity.

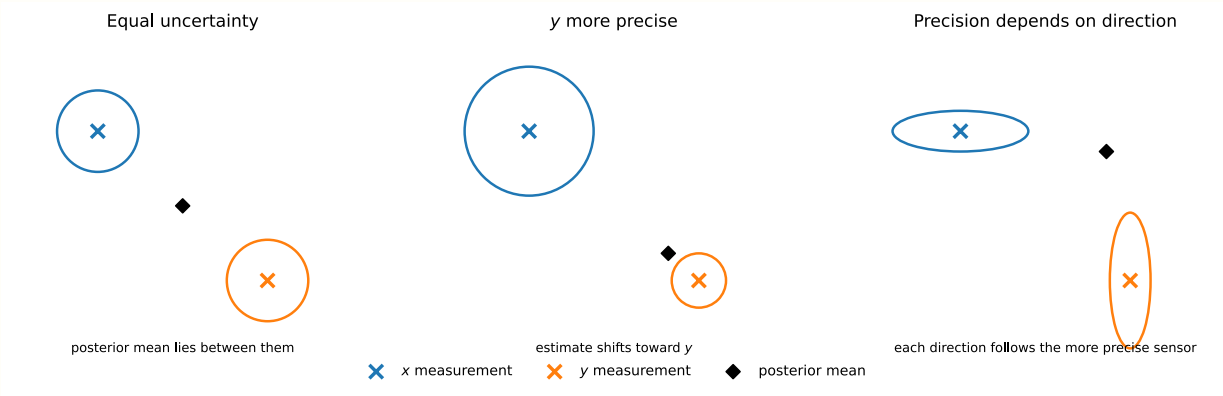
Sensor fusion combines measurements according to their uncertainty

Two noisy measurements of one state

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)\; x \mid z \sim \mathcal{N}(z, \Sigma_x), \quad y \mid z \sim \mathcal{N}(z, \Sigma_y)$$

Same two operations

$$p(z \mid x, y) \quad \rightarrow \quad E[z \mid x, y]$$



Course schematic after Murphy, Book 2 §2.3.2.3, Fig. 2.10; `sensor_fusion_figure.ipynb`

- Equal uncertainty puts the posterior mean between the measurements
- A more precise sensor pulls the posterior mean closer to its measurement
- With anisotropic uncertainty, the weighting can differ by direction